ENGS 104

Optimization Methods for Engineering Applications

Fall 2026

DARTMOUTH ENGINEERING THAYER SCHOOL

For more information and for required policy statements see the DCAL Syllabus Guide.

Course Instructors: Prof. Bryce L. Ferguson (he/him)
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Office Hours: Prof. Ferguson:
Tuesdays 11:00 am - 12:00 pm & Fridays 2:00 pm - 3:00 pm in MacLean 309

Prof. Fugenshuh
Wednesdays 2:00 pm - 3:00 pm & Thursdays 2:00 pm - 3:00 pm in MacLean 307

Staff: Sarah Wall
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Teaching Assistants: Yuwei Hu
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Office Hourse: Wednesday 4:00 pm - 5:00 pm in …

Course Description: An introduction to various methods of optimization and their uses in modern engineering. Students will learn to formulate and analyze optimization problems and apply optimization techniques in addition to learning the basic mathematical principles on which these techniques are based. Topic coverage includes linear programming, nonlinear programming, dynamic programming, combinatorial optimization and Monte Carlo methods.

Goals and Rationale: Teach students the fundamental theory of optimization and provide them with knowledge and computational tools to solve linear, mixed-integer, and convex programs. This course will introduce or refine students' understanding of the underlying optimization principles used in operations research, control, signal processing, and machine learning.

Learning Objectives: Upon completing this course, students will be able to:

1. Describe how to take an optimization problem from the real world and transcribe it as a linear or convex programming problem, identifying the decision variables, objective, constraints, and the set of problems to which it belongs.
2. Apply optimization solvers including the simplex method, gradient descent, and primal-dual algorithms to generate the solution to linear and convex programming problems.

3. Explain the connection between optimization theory and other areas of engineering, including estimation, function approximation, machine learning, control, game theory, and more.
4. Derive the dual optimization problem and appreciate the interpretation of dual variables in various contexts, e.g., constraint penalties, shadow prices, and solution sensitivity.

Course Time: **(3B)** Tu/Th 4:30 - 6:20 pm + F 4:35 - 5:25 pm

Course Location: MacLean 201

Prerequisites: MATH 22 and ENGS 27 or equivalents, or permission of instructor

Teaching Approach: Class time will primarily be used for lecture and discussion on course material. In addition, several class sessions will be devoted to interactive tutorials on using software and implementing material discussed in lecture.

Assessments: Students will complete 4 at home assignments with a mix of mathematical and computational problem solving through the first 8 weeks of the course, and present (in groups of two or three) a research article or proposed approach using or solving optimization programs.

Grading: Homework - 50%
Final Project - 50%

Late homework will receive a 10% reduction in points for each day it is late by. Final presentations are mandatory and must be presented during one of the class meeting times in week 10. The instructor reserves the right to reduce a students grade in the case of numerous, unexcused absences.

Class Resources: Boyd, Stephen, and Lieven Vandenberghe. Convex optimization. Cambridge University Press, 2004. [Available Online]

Vasek Chvatal, Linear Programming, W.H. Freeman, 1983. [Recommended]

Laurence Wolsey, Integer Programming, 2nd edition, Wiley, 2020. [Recommended]

Canvas: Lecture videos, slides, and notes. Homework and project assignment description and submission. http://canvas.dartmouth.edu.

Academic Honor Principle: Students are expected to follow Dartmouth's Academic Honor Principle and Thayer's Academic Honor Policy.

Religious Observances: Dartmouth has a deep commitment to support students' religious observances and diverse faith practices. Some students may wish to take part in religious observances that occur during this academic term. If you have a religious observance that conflicts with your participation in the course, please meet with me as soon as possible—before the end of the

second week of the term at the latest—to discuss appropriate course adjustments.

Student Accessibility and Accommodations: Students requesting disability-related accommodations and services for this course are required to register with Student Accessibility Services (SAS; Apply for Services webpage; student.accessibility.services@dartmouth.edu; 1-603-646-9900) and to request that an accommodation email be sent to me in advance of the need for an accommodation. Then, students should schedule a follow-up meeting with me to determine relevant details such as what role SAS or its Testing Center may play in accommodation implementation. This process works best for everyone when completed as early in the quarter as possible. If students have questions about whether they are eligible for accommodations or have concerns about the implementation of their accommodations, they should contact the SAS office. All inquiries and discussions will remain confidential.

Tentative Schedule

Week or Unit	Topics	Deadlines and Notes
1	Linear Programs, Standard Simplex Method Duality Theory Dual Simplex Method	HW1 assigned
2	Game Theory (Zero-Sum Matrix Games) Revised Simplex Method (Primal and Dual) Mixed-Integer Programming, Branch-and-Bound	
3	Cutting Plane Methods I Cutting Plane Methods II Decomposition Methods I (Lagrange)	HW2 assigned HW1 due
4	Decomposition Methods II (Benders, Dantzig-Wolfe) Stochastic Optimization TA Session - Multi-Variable Calculus Review	
5	Nonlinear Optimization Problems Convex Functions and Convex Sets TA Session - CVXPY	HW3 assigned HW2 due
6	Classes of Convex Programs Robust Optimization and Conic Programming Project Proposal Discussions	
7	Duality I Duality II TA Session - Solver Tools	HW4 assigned HW3 due
8	Algorithms & Solvers I Algorithms & Solvers II Distributed Optimization	
9	Applications: Estimation, Regression, and SVM Applications: Optimal Control	HW4 due

	Applications: Network Routing and Population Games	
10	Final Presentations	Final Reports due